

Problems and Solutions in Woolworths' Online Business Model

DMV302 – Data Mining and Visualization

Assessment 1 - Data Mining Case Study and Data Cleaning

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Executive Summary

1. Purpose of the Report

The purpose of this report is to analyze the impact on Woolworths of the shift in consumer purchasing patterns from in-person to online purchases, such as Amazon and eBay, as a result of COVID-19. It is also the purpose of this report to identify specific challenges and opportunities for Woolworths' business model and strategy.

2. Problem

The impact of COVID-19 has brought unprecedented challenges to traditional retailers. On the other hand, online shopping, which allows customers to make purchases on the spot without human contact, has experienced tremendous growth. Woolworths must adapt to these changes for both online and offline retailers. The question is to identify the specific challenges Woolworths faces and how they affect the business.

3. Main Issues and Research Methods

The report focuses on strengthening Woolworths' online business model, customer acquisition and retention, regulatory measures to uncover and prevent mislabeling and misrepresentation of products, and developing more efficient inventory management strategies. These issues require the use of multifaceted research to analyze market trends, track and analyze consumer purchasing behavior, and utilize data mining techniques and machine learning to anticipate changing consumer needs and develop business strategies.

4. Recommendations

Woolworths is encouraged to increase its online business by tailoring promotions to its needs

and attracting and retaining new and existing customers. Woolworths also recommends strengthening regulatory measures to ensure the authenticity and quality of products and increase consumer confidence and peace of mind. Finally, data mining and predictive analytics should be used to apply inventory management based on fluctuating consumer needs to solve the problems of over-ordering and under-stocking. That are three recommendations for this sammary.

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Introduction

Our target company, Woolworths New Zealand, is a grocery store with 185 stores nationwide, and the impact of the 2020 COVID-19 pandemic has resulted in a shift in consumer purchasing behavior. Woolworths similarly focused its efforts on online shopping instead of offline shopping, where people had to visit in person to purchase products. Woolworths has adopted a consumer-sensitive business model that allows business owners, entrepreneurs, and others to open their own online stores and sell their products and services to customers. The purpose of this study was to increase customer satisfaction with online shopping. The goal is to retain

existing customers and attract new ones, which requires a multifaceted improvement of the business model. The study aims to provide Woolworths with transparency by clarifying the company's approach to improvement. The premise of the study is to investigate improvements to the online shopping model and the use of data mining techniques to improve the business model.

Enhancing Customer Engagement by Data-Driven Personalization

To enhance customer satisfaction with online shopping Woolworths needs to increase immersion more efficiently, resonating with data-driven, more personalized individual preferences. We recommend the use of data mining techniques clustering, anomaly detection, association rule mining, text mining, anomaly detection, and time series analysis to create deeper and more loyal connections with customers.

Using clustering to reveal customer preference

The ability to identify and predict customer behavior is the cornerstone of personalized customer service. Woolworths is currently using Quantum's personalization engine, which allows customers to scan their rewards cards with their purchases to ensure that they get what they want, when they want it. (Tonkin, *Woolworths Ups Stake in Data Analytics*) In this model, customer purchase history, browsing records, frequency, and price range are analyzed using clustering technology to segment customers based on their preferences and provide products that meet their needs.

Personalized Product Recommendations and Discounts

Future service enhancements will use similar clustering techniques to personalize product recommendations and discounts based on customers' purchasing trends and needs, based on their frequency of purchases and purchasing history, thereby increasing their willingness to purchase at Woolworth's and enabling ongoing customer retention. "Single product discounts are "more influential than bundled discounts such as set discounts" in motivating consumers to

purchase products. " (*How'd that get in my cart? how recommendations and Discounts Affect Online Shoppers' trust and purchase decisions*) This is a case in point. Therefore, you need to know your customers better, because you need to pinpoint the products to which you are offering discounts to them.

This service requires tracking a customer's purchase history throughout the year. Therefore, it is important to present personalized advertisements that incorporate purchase and other factors such as season, purchase frequency, temperature to respond to changes in purchases due to temperature changes, the length of time between previous purchases and opening the application, and how soon the same product is purchased again.

Enhancing Regulatory Compliance and Fraud Detection

Personalized discounts are very attractive to customers, but at the same time, it is very important to have a trusting relationship with them. Excessive frequency of discounts may discourage customers from purchasing or may even damage the branding of the product. For example, customers may only purchase products when discounts are offered, or they may question the value of the product itself because of the discount.

Optimizing Discounts Through Association Analysis

In fact, a fixed and continuous discounting strategy for all customers and products leads to a reduced likelihood of success. (Khadivar & Hamed, 2022) Using association analysis to solve this problem requires Woolworths to develop an understanding of the relationship between products, customer behavior, and discounts. Association analysis is based on assumptions and consequences. Therefore, until a certain amount of data is collected, it is necessary to operate on a trial basis and adjust the frequency of discounts as data is collected.

Identifying the frequency of personalized discounts through association analysis requires a wider variety of data to identify customer purchasing behavior, including hidden relationships between data. For example, it is necessary to analyze a wide variety of data such as time of day, frequency of purchase including both with and without discounts, one time purchase amount, number of products purchased, and time spent in the application to identify the intent of the application specifications.

Maintaining quality through inventory control

As well as specifying the frequency of discounts, it is also very important not to lose the added value of the product. In particular, food products have an expiration date, which means that there is a time limit before the product can be sold. One way to avoid this situation is to sell the product at a discount just before the expiration date, but it is also important to sell the product while there is still time left in the inventory to reach the expiration date.

Enhancing Sales with Predictive Analytics

Predictive analytics makes it possible to advertise products based on historical data patterns. Predictive analytics can help identify patterns in past customer data and buying behavior to present personalized products to customers while there is still time to sell them. For this purpose, Woolworths recommends that each product be assigned an ID and that each product's expiration date be centrally managed. For this predictive analysis, he recommends a model that can handle complex data sets, perhaps a random forest regression. Woolworths recommends using data on food products, such as type of food, number of days since harvest, temperature, humidity, etc., as well as customer purchase history and purchasing behavior.

Conclusion

In conclusion, customer behavior and personalized service are key to improving customer engagement at Woolworths. Providing personalized service requires analysis of customers'

shopping and browsing history. Within the service offering, inventory management and related analysis of customer buying behavior are important to maintain product quality. This allows personalized marketing strategies to improve Woolworths' performance. However, we recognize the limitations of excessive discounting and its impact on the brand, as well as the need to adjust the frequency of customization. In the future, further personalization strategies using AI and machine learning technologies should be considered. These elements are essential for Woolworths to maintain its competitive advantage.

Recomendation

Woolworths already has a personalized strategy in place in cooperation with Quantum. Therefore, the recommendations in this section are based on a deeper dive into that service. Based on the above, we recommend that Woolworths advertise personalized products to each of its customers and offer discounts on those products. The goal is to provide existing customers with an experience that goes beyond the purchase each time they use the service, and to make it a strong advantage for new customers to shop at Woolworths. This service will be accomplished using three types of analysis: association analysis to identify the frequency of discounts and advertisements for each customer and product using clustering analysis; and finally, predictive analysis to ensure the quality of the product and to optimize the inventory. These three analyses will provide Woolworths with a model for increasing customer engagement in online shopping.

Reference

Tonkin, C. (n.d.). *Woolworths Ups Stake in Data Analytics*. Information Age.

<https://ia.acs.org.au/article/2021/woolworths-ups-stake-in-data-analytics.html>

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College | University of Arkansas. (n.d.). <https://walton.uark.edu/insights/posts/howd-that-get-in-my-cart-how-recommendations-and-discounts-affect-online-shoppers-trust-and-purchase-decisions.php>

Khadivar, A., & Hamed, P. (2022, October 20). *Providing synthetic data mining model using association rules and clustering for determining discounting strategy (case study: Pegah distribution co.)*. SSRN.
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4206539